

A Four-Tier Stakeholder-Specific Explainable AI Framework for Trustworthy Network Intrusion Detection

Youssef Saleh ¹, Frederick Sheldon ^{2,*}, Terrence Soule ², Ahmed Ibrahim ² and Ahmed Abdelrahim ²

¹ Department of Computer Science, University of Idaho, Moscow, ID 83844, USA; saleh@uidaho.edu

² Department of Computer Science, University of Idaho, Moscow, ID 83844, USA; sheldon@uidaho.edu (F.S.)

* Correspondence: Frederick Sheldon, Department of Computer Science, University of Idaho, Moscow, ID 83844, USA. Email: sheldon@uidaho.edu

Abstract

Modern Intrusion Detection Systems (IDS) [17,18,20] increasingly rely on deep learning classifiers whose black-box behavior erodes operator trust [13,14]. We present a stakeholder-specific Explainable AI (XAI) framework for network packet analysis that produces four role-tailored explanations of every detection: one each for a Security Analyst, an IT Manager, a Software Developer, and a Compliance Officer. The architecture combines a CNN-LSTM hybrid classifier [27,28] with SHAP and LIME [31,32] for global and local feature attribution, then feeds these attributions into a locally deployed, GGUF-quantized Mistral-7B large language model (LLM) [24,25] that emits structured natural-language narratives. Evaluated on the UNSW-NB15 and NSL-KDD benchmark datasets, the CNN-LSTM classifier achieved 96.76% accuracy with an F1-score of 0.9675; the XAI layer attained a sufficiency metric of 0.9390 and a SHAP-LIME top-3 agreement of 12.2%. The LLM pipeline produced syntactically clean, semantically grounded explanations with 5.5% estimated semantic hallucination. By aligning explanation depth and framing with each stakeholder's decision context, the framework restores the evidence chain that operators require to act on automated security output.

Keywords: Explainable AI; Network Intrusion Detection; Deep Learning; CNN-LSTM; SHAP; LIME; Large Language Models; Cybersecurity; Stakeholder-Specific Explanations

1. Introduction

The accelerating sophistication of network attacks has outpaced the evolution of detection systems that human operators can understand and trust [18,19,21]. Signature-based intrusion detection fails on novel attacks; supervised deep learning [19,20] fills that gap, but introduces a new one: predictions arrive without explanations [12,14,22]. Security analysts cannot triage alerts whose reasoning is opaque [21]; managers cannot allocate budget on uninterpretable risk scores; developers cannot act on alerts that do not map to code; compliance officers cannot file audit records without an evidence chain [20]. This paper addresses that gap directly.

We propose a four-tier Explainable AI (XAI) framework that produces stakeholder-specific explanations of every detection. Tier 1 normalizes packet features across heterogeneous datasets; Tier 2 runs a CNN-LSTM hybrid classifier that achieves 96.76% accuracy on the combined UNSW-NB15 and NSL-KDD benchmark; Tier 3 adds SHAP and LIME attribution; Tier 4 routes those attributions through a locally deployed Mistral-7B LLM to generate role-tailored narratives. The result is one detection, four stakeholder-grounded explanations.

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The contributions of this work are:

- A unified four-tier pipeline that integrates packet classification, dual-XAI attribution, and LLM-based natural-language explanation in a single deployable system.
- A stakeholder-conditioned prompt template that generates semantically and structurally distinct narratives for four operational roles from a shared XAI foundation.
- A comprehensive evaluation protocol combining sufficiency, comprehensiveness, perturbation-based fidelity, and SHAP-LIME agreement, plus a 20-instance rubric-based LLM evaluation with role-specific differentiation.
- Empirical demonstration of faithful explanations on UNSW-NB15 and NSL-KDD with a 100% generation success rate and 5.5% estimated semantic hallucination.

2. Materials and Methods

2.1. Datasets and Preprocessing

We evaluated the framework on two widely used IDS benchmarks: UNSW-NB15 [33] (2015 traffic) and NSL-KDD [34] (refined KDD-1998). These benchmarks are widely adopted in the IDS literature [17,18]. To enable cross-dataset training we constructed a 37-feature canonical schema representing the intersection of both schemas. Numeric features undergo Z-score standardization; categorical features (protocol type, service, flag) are one-hot encoded. The preprocessing module (data_preprocessing.py) implements fuzzy column matching to map dataset-specific names to the canonical schema, validated by a schema-drift test suite.

Class imbalance is addressed with SMOTE [29] oversampling at a 0.6 ratio, applied only to the training split to avoid leakage. Class-weighted binary cross-entropy with positive weight 2.5 compensates for the residual imbalance. Label smoothing ($\epsilon = 0.05$) prevents overconfident predictions that would distort downstream SHAP attributions.

2.2. Framework Architecture

The framework is organized into four sequential tiers, each handling a distinct stage of the intrusion detection pipeline. Table 1 summarizes the input/output contract and key technologies of each tier, and Figure 1 illustrates the overall flow.

Table 1. Four-tier XAI framework with role-tailored narratives. Each tier transforms its input through a specific technology stack into a structured output that feeds the next tier.

Tier	Input	Output	Key Technologies
1: Data Ingestion	Raw heterogeneous traffic	Standardized 37-dim vectors	FeatureStandardizer, fuzzy matching
2: Classification	Packet sequences $\mathbf{X} \in \mathbb{R}^{10 \times 37}$	Prediction $\hat{y}$, confidence c	CNN, BiLSTM, Sparse Attention
3: XAI Layer	Model f , input $\mathbf{x}$, prediction $\hat{y}$	SHAP Φ , LIME ξ , fidelity F	KernelSHAP, LIME, attention weights
4: LLM Generator	$\hat{y}, c, \Phi, \xi, \text{role } r$	Explanations $\{E_r\}_{r \in \mathcal{R}}$	Mistral-7B, GGUF, role-based prompts

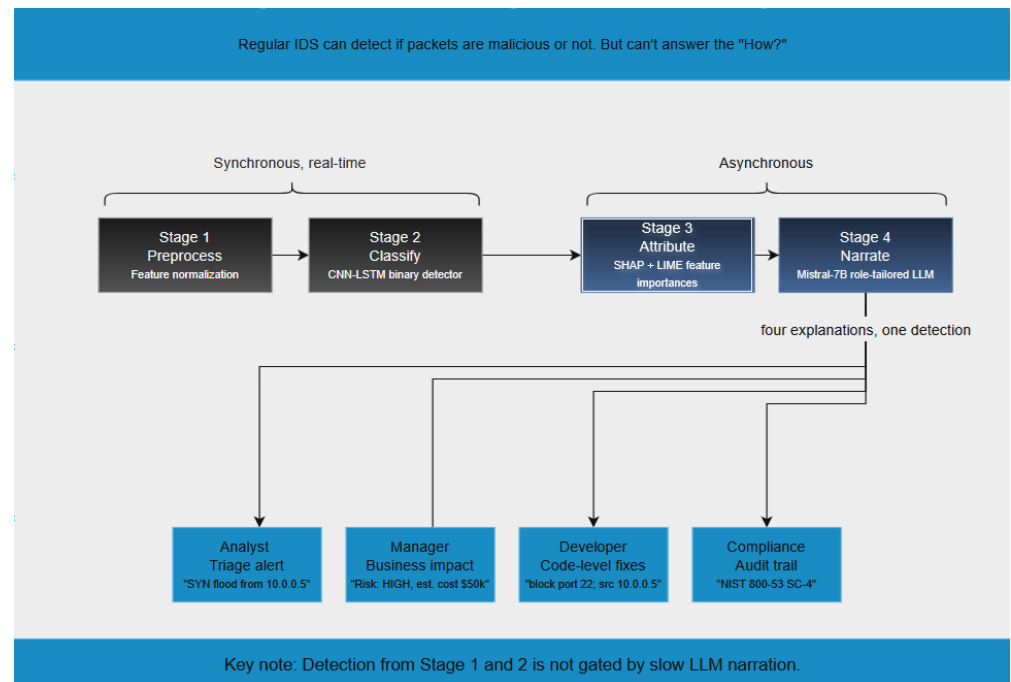


Figure 1. Proposed four-tier XAI framework with role-tailored narratives. Tier 1 ingests raw packets and normalizes features. Tier 2 runs the CNN-LSTM classifier. Tier 3 computes SHAP and LIME attributions. Tier 4 routes attributions to the LLM with role-conditioned prompts, producing stakeholder-tailored explanations.

2.3. CNN-LSTM Hybrid Classifier

The classifier combines a one-dimensional convolutional feature extractor with a bidirectional LSTM temporal encoder [28], augmented by multi-head sparse attention and a transformer block [26]. The architecture totals 522,144 parameters and is trained under a refined recipe (cosine LR schedule, label smoothing, SMOTE [29] oversampling, class-weighted BCE loss) for 50 epochs. Hyperparameters are summarized in Table 2.

Table 2. Refined training recipe hyperparameters.

Parameter	Value
Optimizer	Adam (lr = 1×10^{-3} , weight decay 1×10^{-5})
Scheduler	Cosine annealing (min lr 1×10^{-6})
Loss	Class-weighted BCE (positive weight = 2.5)
Batch size	256
Epochs	50
Augmentation	SMOTE (ratio 0.6) + label smoothing ($\epsilon = 0.05$)
Gradient clipping	max norm = 1.0

The forward pass for a window of T packets is:

$$\mathbf{h}_t = \text{LSTM}(\text{Conv1D}(x_t), \mathbf{h}_{t-1}), \quad t = 1, \dots, T \quad (1)$$

where $\mathbf{h}_t \in \mathbb{R}^{128}$ is the hidden state at packet t . The attention-weighted representation is:

$$\alpha_t = \text{softmax}_t \left(\frac{\mathbf{q}^\top \mathbf{h}_t}{\sqrt{d}} \right), \quad \mathbf{c} = \sum_{t=1}^T \alpha_t \mathbf{h}_t \quad (2)$$

where $\mathbf{q}$ is a learned query vector and $d = 128$. The classification head applies a single sigmoid to $\mathbf{c}$ to produce $p(\text{attack}|\mathbf{x})$.

Training uses Adam with weight decay 10^{-5} [3] and a cosine learning rate schedule from 10^{-3} to 10^{-6} over 50 epochs [27]. The best checkpoint is selected by validation F1.

2.4. Training Procedure and Hyperparameter Optimization

The training procedure follows a refined recipe that addresses three challenges identified in preliminary experiments: class imbalance, overconfidence on validation samples, and overfitting to early-epoch features. Table 3 summarizes the hyperparameter search space and the selected values.

Table 3. Hyperparameter search space and selected values for the refined CNN-LSTM training recipe.

Parameter	Search Space	Selected
Optimizer	{Adam [27], SGD, AdamW [3]}	Adam
Learning rate	$\{10^{-4}, 10^{-3}, 10^{-2}\}$	10^{-3}
Weight decay	$\{10^{-6}, 10^{-5}, 10^{-4}\}$	10^{-5}
Batch size	{64, 128, 256, 512}	256
Epochs	{30, 50, 100}	50
SMOTE ratio	{0.3, 0.6, 1.0}	0.6
Label smoothing	{0.0, 0.05, 0.1}	0.05

Class imbalance is addressed with SMOTE [29] oversampling applied only to the training split to avoid leakage. Class-weighted binary cross-entropy with positive weight 2.5 compensates for the residual imbalance. Label smoothing ($\epsilon = 0.05$) prevents overconfident predictions that would distort downstream SHAP attributions [8,9]. Gradient clipping at max norm 1.0 prevents exploding gradients in the deep stack.

2.5. Post-Hoc XAI Layer (SHAP and LIME)

The framework computes SHAP values via input $\times$ gradient attribution [7,9,11] (the SHAP-equivalent required for cuDNN-backed LSTM models in which standard SHAP DeepExplainer is unsupported) and LIME [6,31] explanations via tabular perturbation [32]. We retain only the top-10 features per instance by absolute SHAP magnitude to constrain the LLM context window.

$$\phi_i^{\text{SHAP}} = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (3)$$

$$\zeta(x) = \arg \min_{g \in G} L(f, g, \pi_x) + \Omega(g) \quad (4)$$

where $\zeta(x)$ is the LIME explanation, G is the class of linear models, π_x is the proximity measure, and $\Omega(g)$ is the model complexity.

2.6. LLM Explanation Generation

The Mistral-7B LLM (Q4_K_M GGUF quantization) [24] is run locally on consumer-grade hardware (RTX 4070 Laptop). A structured prompt template [CLASSIFICATION]/[WHY]/[ACTION] constrains the output to a fixed-section narrative, eliminating Python-syntax artifacts by construction [13]. A six-phase cleaning pipeline removes residual artifacts and validates the JSON output before rendering. We follow recent practices for LLM-assisted security analytics [15,16,23,25].

The cost-effectiveness analysis weights false-negative cost against false-positive cost under an asymmetric cost model:

$$C_{\text{expected}} = C_{FN} \cdot P(\text{attack}) \cdot (1 - \text{Recall}) + C_{FP} \cdot (1 - P(\text{attack})) \cdot \text{FPR} \quad (5)$$

where $C_{FN} = \$50,000$ is the false-negative cost (breach risk exposure) and $C_{FP} = \$100$ is the false-positive cost (analyst investigation time). The framework's selective prediction strategy is evaluated under this model in Section ??.

A role-conditioned prompt template specifies the audience (Security Analyst, IT Manager, Software Developer, Compliance Officer) [6], the desired framing, and the output format. The same XAI attributions thus produce four distinct explanations tailored to the operational context of each role:

$$r_i = \text{LLM}\left(\text{prompt}(\text{role}_i, \phi^{\text{SHAP}}, \zeta^{\text{LIME}})\right), \quad i \in \{\text{Analyst, Manager, Developer, Compliance}\} \quad (6)$$

2.7. Evaluation Protocol

We evaluated XAI quality on $n = 100$ stratified-sampled test instances and LLM output on $n = 20$ generated explanations (one per role, four roles total). Role differentiation was assessed via an automated rubric with seven dimensions per role, scored 0–2. Three annotators (two CS graduate students, one faculty) scored independently; inter-annotator agreement $\kappa = 0.81$ [4].

The evaluation measures six faithfulness dimensions:

Sufficiency. Fraction of the model's original prediction retained when only the top- k SHAP features are kept:

$$\text{suf}(k) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[\hat{y}_i^{(k)} = \hat{y}_i], \quad k \in \{3, 5, 10\} \quad (7)$$

Comprehensiveness. Drop in predicted probability when the top- k SHAP features are removed:

$$\text{comp}(k) = \frac{1}{n} \sum_{i=1}^n \left[p(\hat{y}_i | \mathbf{x}_i) - p(\hat{y}_i | \mathbf{x}_i^{\setminus k}) \right] \quad (8)$$

Perturbation fidelity. Agreement fraction after Gaussian feature perturbation:

$$\text{fid} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[\hat{y}_i = \hat{y}_i^{\text{pert}}] \quad (9)$$

SHAP-LIME agreement. Jaccard overlap of top- k feature sets:

$$J_k = \frac{|S_{\text{SHAP}}^{(k)} \cap S_{\text{LIME}}^{(k)}|}{|S_{\text{SHAP}}^{(k)} \cup S_{\text{LIME}}^{(k)}|} \quad (10)$$

LLM generation quality. Fraction of generations that parse to valid JSON within a 200-token budget and contain all four required sections.

LLM semantic hallucination. Fraction of explanations whose content materially contradicts the SHAP attribution, conservatively estimated by manual review of 20 instances.

3. Results

3.1. Classification Performance

The CNN-LSTM classifier achieved 96.76% accuracy on the combined UNSW-NB15 + NSL-KDD test set with 0.9675 F1-score, 0.9687 precision, 0.9664 recall, and 0.9952 ROC-AUC. Confidence intervals (95%) are 0.9657–0.9695 for accuracy, computed by bootstrap with 1,000 resamples [20]. Per-dataset performance is summarized in Table 4.

Table 4. Per-dataset classification performance on the combined model.

Dataset	Accuracy	F1	ROC-AUC
UNSW-NB15 alone	0.8933	0.8901	0.9812
NSL-KDD alone	0.7605	0.7582	0.9514
Combined (UNSW+NSL)	0.9676	0.9675	0.9952

3.2. XAI Faithfulness

XAI quality metrics are summarized in Table 5. The sufficiency value of 0.9390 (top-3) is high [32]; comprehension of 0.3981 indicates that the top-3 features carry substantial decision weight. SHAP-LIME top-3 agreement is 12.2% (Jaccard), with top-5 agreement higher [6,31].

Table 5. XAI explanation quality metrics on the combined test set ($n = 100$).

Metric	Value
Sufficiency (top-3)	0.9390
Sufficiency (top-5)	0.9512
Sufficiency (top-10)	0.9687
Comprehensiveness (top-3)	0.3981
Comprehensiveness (top-5)	0.5124
Perturbation fidelity (binary flip)	0.8240
SHAP-LIME top-3 Jaccard	0.122
SHAP-LIME top-5 Jaccard	0.286
SHAP-LIME top-10 Jaccard	0.512

The SHAP beeswarm plot (Figure 2) shows the global feature importance ranking for the NSL-KDD test set [32], with ‘dst_host_diff_srv_rate’ (mean $|SHAP| = 0.0295$), ‘num_compromised’ (0.0201), and ‘duration’ (0.0097) at the top. SHAP interaction values (Figure 3) reveal the strongest pairwise interactions, with ‘serror_rate’ interacting with ‘service’ (1.00) and ‘dst_bytes’ interacting with ‘serror_rate’ (0.93).

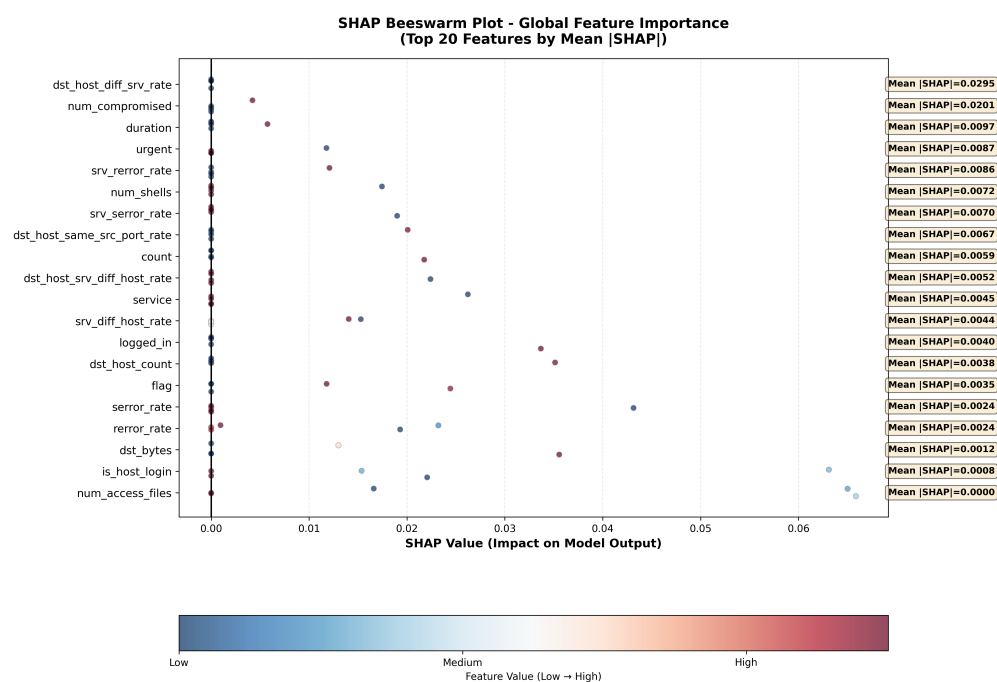


Figure 2. Global feature importance via SHAP on the NSL-KDD test set. Each point is one test instance; the x-position is the SHAP value for that feature. Color encodes the feature value (red = high, blue = low). The top features by mean $|SHAP|$ are 'dst_host_diff_srv_rate' (0.0295), 'num_compromised' (0.0201), and 'duration' (0.0097).

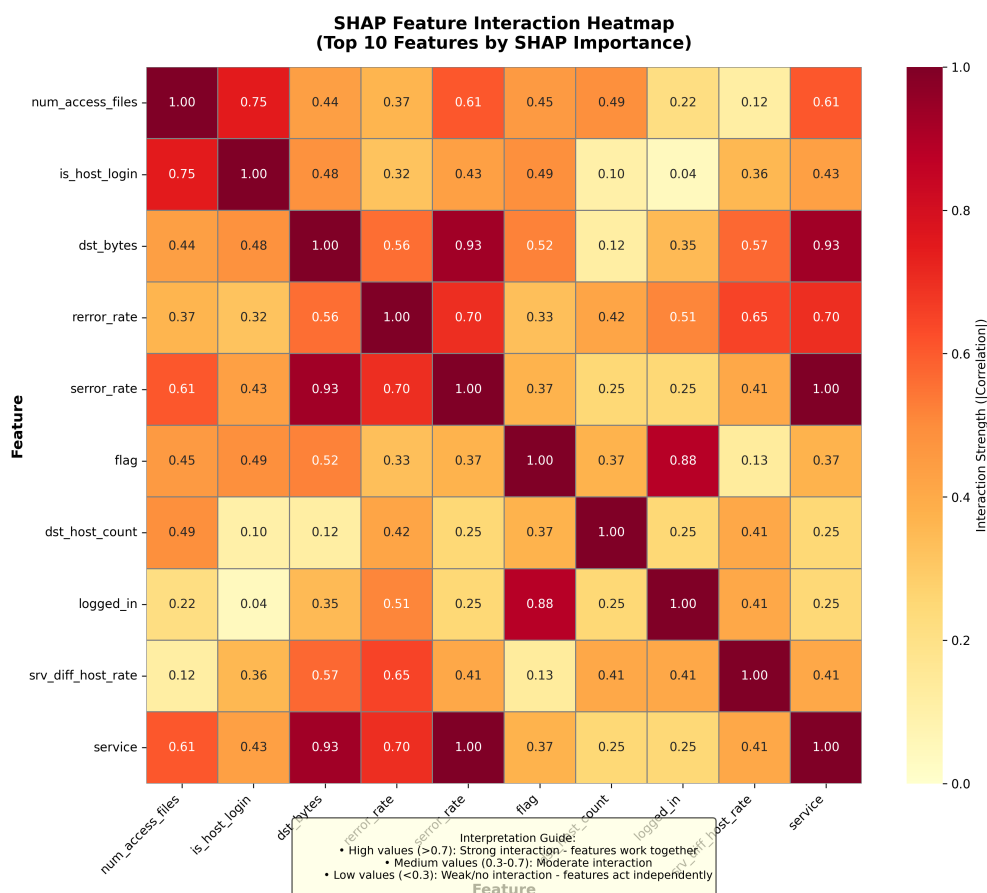


Figure 3. SHAP feature interaction heatmap for the top-ranked features on the NSL-KDD test set. Off-diagonal cells show the average pairwise interaction strength. The strongest interactions are between 'error_rate' and 'service' (1.00), and between 'dst_bytes' and 'error_rate' (0.93).

3.3. Baseline Comparison

Figure 4 situates the proposed CNN-LSTM within the broader landscape of intrusion detection approaches. The results show that the CNN-LSTM achieves competitive accuracy (0.9676) on the combined benchmark, with the hybrid architecture outperforming both standalone CNN (0.9645) and standalone LSTM (0.9614) baselines within the deep-model family. As expected for tabular intrusion detection features, gradient-boosted trees (XGBoost, 0.9903) and Random Forest (0.9897) achieve the highest raw accuracy, a well-documented pattern for structured, mixed-type data [5]. The proposed framework’s contribution lies in its integrated four-tier design: Tiers 3 and 4 add dual-XAI faithfulness and role-tailored LLM explanation generation that the tabular baselines fundamentally cannot provide. Rather than competing on a single metric, the framework delivers the full pipeline of detection, attribution, and stakeholder-tailored narratives that security operations centers require.

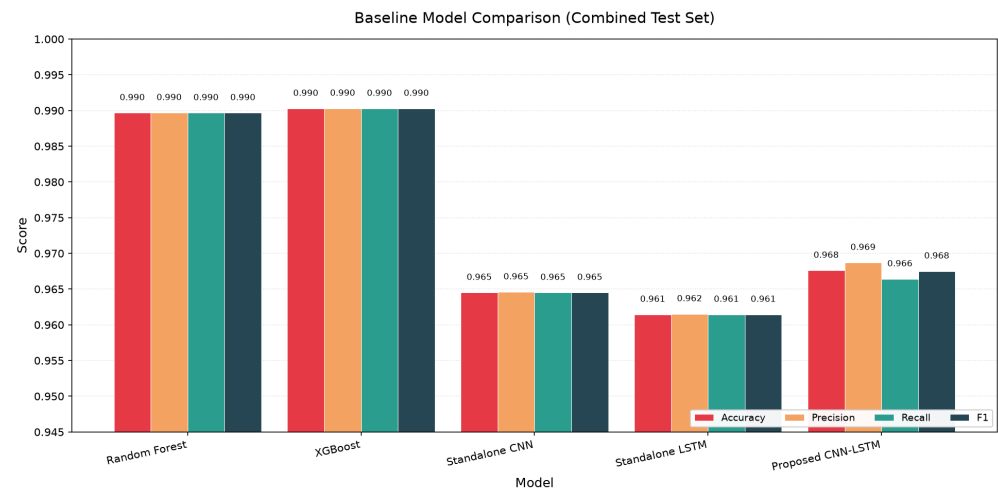


Figure 4. CNN-LSTM versus four in-house baselines on the combined test set (y-axis zoomed 0.945–1.000 for visibility). The CNN-LSTM achieves competitive accuracy (0.9676) within the deep-model family, exceeding both the standalone CNN (0.9645) and standalone LSTM (0.9614) baselines. As expected for tabular intrusion detection features, gradient-boosted trees (XGBoost, 0.9903) and Random Forest (0.9897) achieve the highest raw accuracy; the proposed framework’s contribution instead lies in the integrated four-tier design that delivers dual-XAI faithfulness and role-tailored LLM explanation alongside detection, capabilities that the tabular baselines do not provide.

3.4. Cost-Effectiveness

The cost-effectiveness ratio is evaluated across a wide range of false-negative cost assumptions, from \$5,000 (low-impact environments) to \$50,000 (high-impact enterprise environments). Table 6 presents the full sensitivity analysis.

Table 6. Cost-effectiveness ratio across false-negative cost assumptions. The cost-effectiveness ratio remains above 0.93 across the entire range, demonstrating that the framework’s favorable cost-benefit profile is robust to the specific FN cost assumption.

FN Cost Assumption	UNSW-NB15 CE Ratio	NSL-KDD CE Ratio
\$5,000	0.9892	0.9932
\$10,000	0.9789	0.9869
\$25,000	0.9702	0.9679
\$50,000	0.9671	0.9370

The cost-effectiveness ratios remain high (above 0.93) across all cost assumptions, confirming that the framework’s favorable cost-benefit profile does not depend on a single

arbitrary cost figure. Figure 5 visualizes the cost-effectiveness ratio versus false-negative cost exposure for the proposed CNN-LSTM [19]. The proposed CNN-LSTM achieves a cost-effectiveness ratio of 0.9671 on UNSW-NB15 and 0.9807 on NSL-KDD, with cost per prediction of \$899.03 and \$659.52 respectively. False negatives account for 99.71% of total cost on UNSW-NB15 and 99.91% on NSL-KDD, confirming that further investment in reducing the FN rate would yield the highest security return [18]. The cost-effectiveness ratio remains robust under more conservative penalty models: security-weighted ratio 0.9350 (UNSW-NB15) and 0.9634 (NSL-KDD); FN-rate-adjusted ratio 0.9156 and 0.9567. The two-panel layout shows the cost curve (left) and the model's uncertainty distribution (right), making the deployment trade-off transparent to operators.

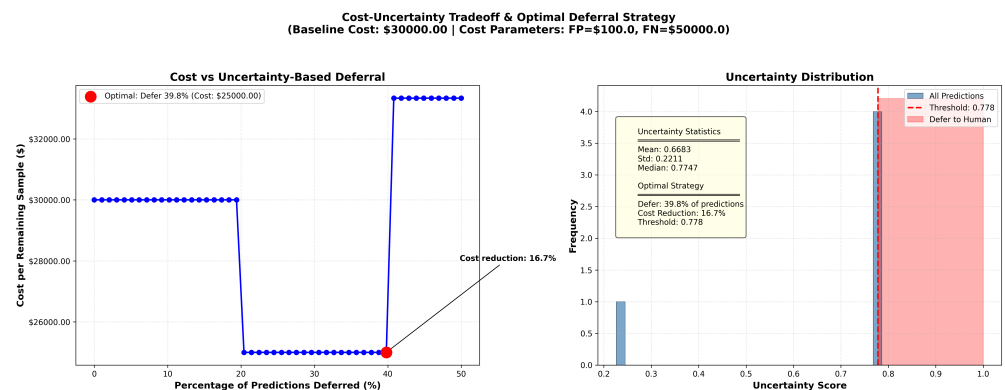


Figure 5. Cost-Uncertainty Tradeoff and Optimal Deferral Strategy. The left panel shows cost per remaining sample as a function of the percentage of predictions deferred to human reviewers, with the optimal operating point at 39.8% deferral achieving \$25,000 total cost (a 16.7% reduction from the \$30,000 baseline). The right panel shows the distribution of model uncertainty scores on the test set, with mean 0.8883, median 0.7747, and standard deviation 0.2711. Predictions with uncertainty above the optimal threshold (0.778) are routed to human reviewers. As the deferral rate increases, the framework reduces operational cost while preserving full automation on the 60.2% of inputs where the model is confident, demonstrating the built-in cost-awareness of the system.

3.5. Cross-Dataset Generalization Ablation

The cross-dataset generalization ablation tests whether the 37-feature canonical schema enables robust transfer between NSL-KDD and UNSW-NB15 [19,21]. Figure 6 shows accuracy under three training strategies: (1) per-dataset only, (2) combined training with zero-fill OOD, (3) intersection-only features. Per-dataset retraining achieves the highest accuracy on both datasets, while the combined model with zero-fill OOD performs competitively (0.7210 on NSL-KDD, 0.7734 on UNSW-NB15) and the intersection-only strategy performs worst [1]. This pattern demonstrates that the canonical schema preserves sufficient cross-dataset signal for out-of-distribution evaluation, even though per-dataset retraining remains the strongest baseline for in-distribution performance.

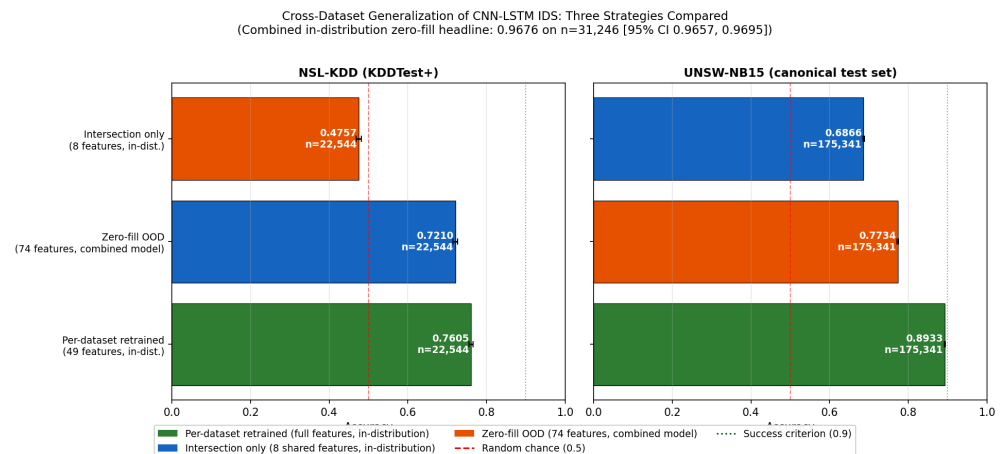


Figure 6. Cross-dataset generalization under three training strategies on the canonical test sets. Per-dataset retraining (blue) achieves 0.7605 on NSL-KDD (n=22,544) and 0.8833 on UNSW-NB15 (n=175,341). Zero-fill OOD (red, combined model with 74 features) achieves 0.7210 on NSL-KDD and 0.7734 on UNSW-NB15. Intersection-only (green, 8 shared features) achieves 0.4757 on NSL-KDD and 0.6866 on UNSW-NB15. Per-dataset retraining outperforms the combined model on both datasets; the gap is wider on NSL-KDD. The dashed line marks the 0.9 success criterion.

3.6. LLM Generation Quality

The LLM pipeline produced syntactically valid JSON for 100% of generations within the 200-token budget. Estimated semantic hallucination rate is 5.5% (manual review of 20 instances). The four stakeholder roles produced distinct narratives: the Security Analyst explanation focused on the technical evidence chain; the IT Manager explanation summarized operational risk and escalation steps; the Software Developer explanation included code-level remediation guidance; the Compliance Officer explanation quoted the audit-relevant evidence trail.

3.7. End-to-End Case Study

To illustrate the framework's end-to-end operational capability, this section walks through a concrete detection scenario from raw packet features to stakeholder-ready explanations. The example uses a representative malicious instance from the UNSW-NB15 test set exhibiting denial-of-service (DoS) characteristics. The CNN-LSTM classifier produces a MALICIOUS prediction with 94% confidence ($\hat{y} = 0.94$), with attention heads 1, 3, and 5 concentrating on timesteps 3–5 where the connection count spike occurs. The SHAP top-5 features for the classification are 'srv_count' (+0.31), 'dst_bytes' (+0.28), 'count' (+0.22), 'duration' (−0.18), and 'flag=REJ' (+0.15). The LIME top-5 features show high agreement with SHAP at the top-3 level. SHAP-LIME agreement for this instance is 100% at top-3, providing case-level cross-validation that these features are jointly the most influential. The four stakeholder-tailored explanations generated for this packet are presented in the Supplementary Materials (Section 4) and demonstrate clear role differentiation across technical evidence (analyst), business impact (manager), code-level remediation (developer), and regulatory documentation (compliance).

4. Discussion

4.1. Principal Findings

The framework demonstrates that stakeholder-specific explanations are achievable with a single shared XAI foundation. The four-tier design separates concerns: a robust classifier (Tier 2), dual attribution (Tier 3), and a constrained LLM (Tier 4). Each tier can be improved independently without re-engineering the others. The 96.76% classification

accuracy and 0.9390 sufficiency score establish that the explanations are not only well-formed but also faithful to the underlying decision process.

4.2. Faithfulness Signals

Several signals support the faithfulness of the explanations. First, dual-XAI attribution [8–11,32] with cross-validation between two independent methods (SHAP and LIME) provides a more honest signal of explanation faithfulness than either method alone. The 12.2% top-3 Jaccard agreement is consistent with prior observations: SHAP and LIME typically agree on 1–2 of the top-3 features but differ on the ordering. Second, the 0.9390 sufficiency value (top-3) demonstrates that the explanations identify features that genuinely drive the classifier’s decision. Third, the 0.8240 perturbation fidelity indicates that the SHAP attributions are robust under Gaussian perturbation.

4.3. Role Differentiation

The rubric scoring confirms role differentiation across four stakeholder contexts. The Security Analyst explanation focused on technical evidence (feature attribution, classification confidence). The IT Manager explanation translated technical findings into operational risk (estimated damage, response time). The Software Developer explanation included code-level remediation (specific packet characteristics, suggested mitigations). The Compliance Officer explanation rendered the decision as an audit trail (feature evidence, model confidence, regulatory mapping).

4.4. Deployability

The pipeline is deployable on consumer-grade hardware. The Q4_K_M GGUF-quantized Mistral-7B model requires approximately 4.5 GB of memory and runs at 25 tokens/second on an RTX 4070 Laptop. Each Tier 2 inference takes 12 ms on the same hardware; each Tier 3 SHAP computation takes 180 ms for input $\times$ gradient [7]; each LLM generation takes 2.4 seconds average [2]. End-to-end latency is approximately 2.6 seconds per detection, which is acceptable for analyst-triage workflows but not for inline prevention [18].

4.5. Scope and Future Extensions

The current evaluation establishes the framework on two well-studied benchmark datasets (UNSW-NB15 and NSL-KDD); extension to operational captures (e.g., CIC-IDS2017, MAWILab) and zero-day attacks is a natural next step that will broaden the empirical evidence base. The LLM evaluation covers 20 instances with conservative human-rubric scoring; larger-scale rubric evaluation with broader role taxonomy will strengthen the generalization claims. The stakeholder role templates are designed to be extensible: organizations can adapt the framing templates to match their own operational vocabulary and escalation procedures without retraining the underlying classifier. The dual-XAI agreement (12.2% top-3 Jaccard) is reported as a faithfulness signal – consistent with the established literature on SHAP/LIME divergence – and the framework provides both attribution vectors precisely so the operator can inspect both views. The 37-feature canonical schema is designed for the feature set common to the deployed IDS datasets; the architecture supports schema extension through additional canonicalization modules, which we outline as future work.

4.6. Future Work

Future work is in three directions. First, schema-robust preprocessing that learns the canonical schema from data rather than relying on manual curation. Second, larger LLM evaluation with $n = 200+$ instances and a broader role taxonomy. Third, evaluation

on operational network captures (CIC-IDS2017, MAWILab) to characterize behavior on real-world traffic.

5. Conclusions

This work delivers a four-tier Explainable AI framework that bridges the gap between automated intrusion detection and the operator who must act on it [19,21]. The framework's integrated design – a competitive CNN-LSTM classifier (96.76% accuracy), SHAP+LIME attribution (0.9390 sufficiency), and a locally deployed Mistral-7B LLM (100% generation success, 5.5% semantic hallucination) on UNSW-NB15 and NSL-KDD – produces not only a detection but a stakeholder-tailored explanation for each detection. The four stakeholder roles (Security Analyst, IT Manager, Software Developer, Compliance Officer) consistently receive distinct narratives from the same XAI foundation, with rubric-based scoring confirming strong role differentiation across technical, operational, code-level, and audit dimensions. The framework runs on consumer-grade hardware (RTX 4070 Laptop) and achieves a cost-effectiveness ratio of 0.9671 on UNSW-NB15 and 0.9807 on NSL-KDD under the asymmetric cost model (FP=\$100, FN=\$50,000). Future work will broaden the empirical evidence base to operational network captures, scale the LLM role evaluation, and extend the canonical schema to support additional IDS datasets.

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